Quiz 4- Winter 2026- Solutions

February 5, 2026 9:31 AM

MATH 3042- Quiz 4

Full Name:

Time Allowed: 30 minutes Set:

Expected response for probability questions: Mathematical representation of the probability in question (e.g. $P(x \geq 7)$) followed by the correct numeric result.

1. (4 points) Match each scenario to the most appropriate probability distribution. No need to calculate the probability. Two or more might match the same distribution.

Scenario	**Options**
a) A quality control inspector randomly selects 15 smartphones from a batch of 100, where 12 are defective. Finding the probability that exactly 3 defective phones are selected. Hypergeometric	(d) Binomial
b) A penetration tester attempts to crack a password. Each attempt has a 5% success rate. Finding the probability it takes exactly 10 attempts to succeed. Geometric	(a) Hypergeometric
c) A database receives queries at an average rate of 20 per minute. Finding the probability of receiving exactly 25 queries in one minute. Poisson	(b) Geometric
d) A machine learning model makes 50 predictions, each with an 85% accuracy rate. Finding the probability of exactly 45 correct predictions. Binomial	(c) Poisson

2. (6 points) A cybersecurity system has a 92% success rate at detecting malware. If 80 files with malware are scanned:

$$X \sim Binomial(n = 80, p = 0.92)$$

a) Calculate the probability that exactly 76 files are identified as having malware. Use the provided probabilities.

$$P(X = 76) = dbinom(76, 80, 0.92) = 0.8911 - 0.7765 = 0.1145$$

b) Is it unusual to correctly identify 54 files as having malware? Use $|z| > 2$ criteria.

$E[X] = np = (80 \times 0.92) = 73.6$

$\sigma = \sqrt{(80 \times 0.92 \times 0.08)} = 2.43$

$z = \dfrac{54 - 73.6}{2.43} = -8.1$

Yes, it is very unusual to correctly identify 54 files or less as having malware.

Cutoff point for $z = -2$: $-2 = \frac{x - 73.6}{2.43} \approx 69$ anything below 69 is unusually low

3. (5 points) A code repository contains 40 commits, of which 9 introduce bugs. A code review randomly selects 12 commits to examine.

a) What is the probability that exactly 1 buggy commit is found? Please evaluate the probability by using a calculator.

$$P(x = 1) = \frac{C(9,1) \times C(31,11)}{C(40,12)} = 13.6\%$$

b) What is the probability that at least 1 buggy commit is found? Please evaluate the probability by using a calculator.

$$P(x = 0) = \frac{C(9,0) \times C(31,12)}{C(40,12)} = 2.53\%$$
$$P(x \geq 1) = 1 - P(x = 0) = 97.47\%$$

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4. (4 points) A debugging process has a 15% chance of identifying the bug on each test run.

a) What is the probability that the bug is found on the 8th test run? Evaluate as a percentage, correct to one digit after the decimal point.

$$P(x = 8) = (0.85)^7(0.15) = 4.8\%$$

b) What is the expected number of test runs needed?

$$E[X] = \frac{1}{p} = \frac{1}{0.15} = 6.7$$

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5 (4 points) A web API receives an average of 15 requests per minute. By using the provided probability values, determine:

b) What is the probability of receiving more than 10 requests in a minute?

$$P(x > 10) = 1 - ppois(10, 15) = 1 - 0.1185 = 88.15\%$$

d) What is the probability of receiving exactly 30 requests in a 2-minute interval?

$$P(x = 30) = dpois(30,30) = 7.26\%$$

Binomial Distribution

Probability Mass Function (PMF):

$$P(X = k) = C(n,k) \times p^{k} \times (q)^{n-k}$$

$$k = 0, 1, 2, \dots, n$$

Mean and Variance:

- $Mean: \mu = np$
- $Variance: \sigma^2 = npq$

Hypergeometric Distribution

Probability Mass Function (PMF):

$$P(X = k) = \frac{[C(K,k) \times C(N-K, n-k)]}{C(N,n)}$$

Mean and Variance:

$$Mean: \mu = n \times \left(\frac{K}{N}\right)$$

$$Variance: \sigma^2 = n \times \left(\frac{K}{N}\right) \times (1 - \frac{K}{N}) \times [\frac{N-n}{N-1}]$$

Geometric Distribution

Probability Mass Function (PMF):

$$P(X = k) = q^{k-1} \times p$$

$$for\ k = 1, 2, 3, \dots$$

Mean and Variance:

$$Mean: \mu = \frac{1}{p}$$

$$Variance: \sigma^2 = \frac{q}{p^2}$$

Poisson Distribution

Probability Mass Function (PMF):

$$P(X = k) = \frac{(\lambda^{k} \times e^{-\lambda})}{k!}$$

$$for\ k = 0, 1, 2, 3, \ldots$$

Mean and Variance:

- $Mean: \mu = \lambda$
- $Variance: \sigma^2 = \lambda$

Provided probability values (not all of these values are used):

pbinom(76, 80, 0.92) = 0.8911

pbinom(75, 80, 0.92) = 0.7765

ppois(9,15) = 0.0699

ppois(10,15) = 0.1185

ppois(11,15) = 0.1848

dpois(30,15) = 0.00022

dpois(30,30) = 0.0726

ppois(30,15) = 0.9998

ppois(30,30) = 0.5483